

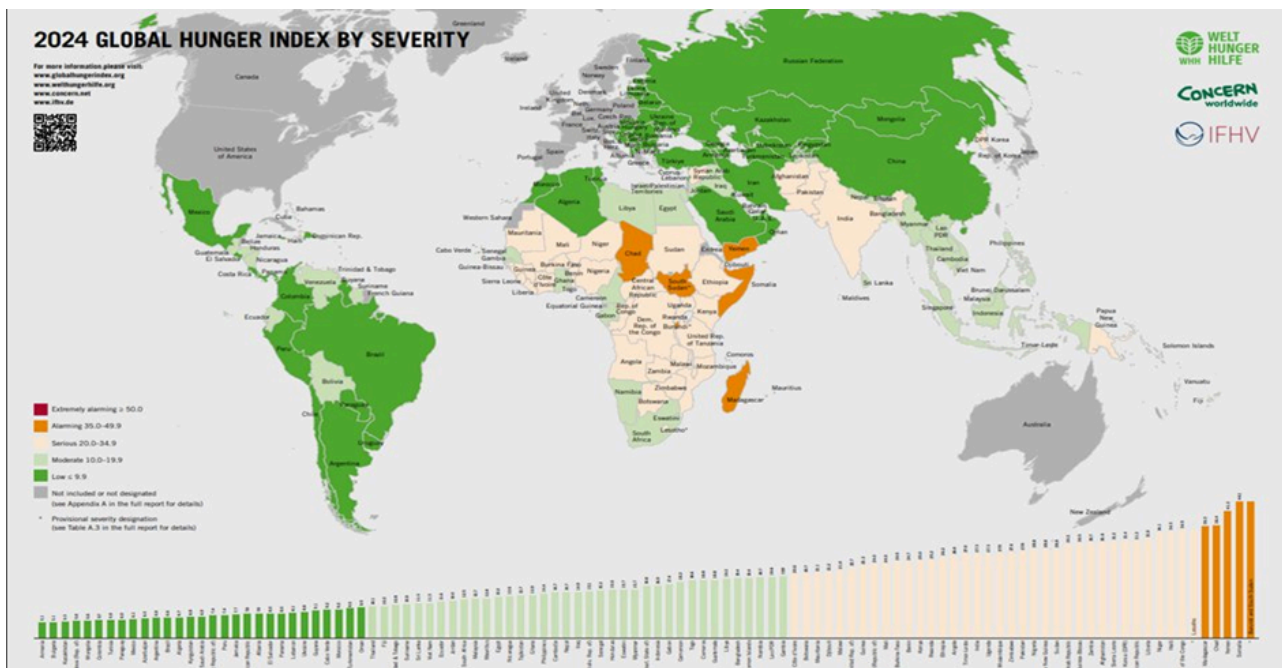
[Cadmus] Assignment 2 - Deconstruct, Reconstruct Web Report

Improving Accessibility, Information Density, and Projection in the 2024 Global Hunger Index Map

Original

Figure 1

2024 Global Hunger Index by Severity (Welthungerhilfe et al., 2024)



Note. Screenshot of the published 2024 Global Hunger Index by Severity poster (Welthungerhilfe et al., 2024). The figure consists of a world choropleth showing severity classification combined with a sorted bar chart of all 127 ranked countries' GHI scores beneath the map. The artefact is the

static, downloadable PDF poster (a Level 1 artefact under Schulmeister, 2003), not the interactive web version of the same data.

Retrieved from:

https://www.welthungerhilfe.org/fileadmin/pictures/publications/en/studies_analysis/2024-global-hunger-index-poster-map.pdf.

Objective and audience

The original visualisation is the headline figure of the 2024 GHI report. It is a composite figure: a world choropleth on top, paired with a horizontal bar chart of all 127 ranked countries underneath. Its objective is communicative rather than analytical: to show readers, at a glance, which countries face the most severe hunger, how that severity is distributed geographically, and where each individual country sits on the 0-100 score range. The map collapses the score into five named severity bands (Low, Moderate, Serious, Alarming, Extremely Alarming) so that a person with no statistical background can read it without having to interpret the score itself. In the 2024 edition, six countries fell into the Alarming band: Somalia, Yemen, Chad, Madagascar, Burundi, and South Sudan (Welthungerhilfe et al., 2024).

The audience is non-specialist and global: policy advocates, journalists, donors, classroom teachers, and members of the general public who follow humanitarian reporting. Several design choices follow from this. Country names are written in full instead of using ISO codes, severity is encoded categorically rather than numerically, the layout is poster-shaped so it can be printed and pinned up, and the figure is published as a stand-alone PDF so it can be shared independently of the report. By Schulmeister's (2003) taxonomy this places the artefact at Level 1, the reader can only view it, not interact with it. The GHI report does also publish an interactive map online, but that is a separate Level 3 artefact and is not the figure I have selected.

Improvements

I propose three improvements. They sit on three different design axes: colour and accessibility, information density, and projection plus typography, so the reconstruction tests a genuine spread of skills rather than rehearsing one fix three times.

1. Use a sequential, colourblind-safe palette and split the single "Not designated" grey into two distinguishable neutrals.

The original uses a five-band palette running from green (Low) through pale green (Moderate) and orange (Alarming) to dark red (Extremely Alarming, though no country falls into this band in 2024). This looks intuitive but is the wrong family of colours: GHI severity is an ordinal sequential scale that goes from less hunger to more hunger, not a diverging scale that goes from good to bad about a meaningful midpoint. Green at the Low end implies a positive valence the data does not have; every country on the scale has some level of hunger. The choice also creates problems for readers with red-green colour vision deficiency, who can struggle to separate Low and Moderate, both of which use green hues. A second issue compounds this. Every country that is either high-income (and so not scored by the GHI) or genuinely missing data is filled with the same medium

grey labelled "Not included or not designated." This collapses two very different things: "known to be food-secure" and "unknown" into a single visual category, and it visually empties most of the Northern Hemisphere into ambiguity. My reconstruction uses a single sequential YlOrRd ramp (Brewer & Harrower, 2013), which preserves ordinal ordering, separates cleanly under deutan, protan, and tritan vision simulation (verified using the colorblindr R package), and reads correctly when printed in greyscale. The two non-scored cases are then split: a warm off-white for "High-income, not scored by GHI" and a cool grey for "No data / insufficient." A further benefit of the sequential ramp is that it partially offsets the well-known area-bias of choropleth maps (Roth, 2017), where large countries draw the eye regardless of their score; a darker, more saturated red on a small alarming country like Burundi now reads more strongly against the surrounding lighter fills than the original's pale orange did.

2. Direct-label the Alarming countries and add a small adjacent panel showing the largest reductions in GHI score since 2000.

The original is actually a composite figure: a world choropleth on top, and a horizontal bar chart of all 127 ranked countries underneath, sorted by severity colour and then by score within each band. The scores are therefore visible, but they are visible in the wrong place. To answer the question "which Alarming country has which score," a reader has to find the orange-shaded country on the map, then scan a long list of small rotated country labels along the bottom of the figure to match it to a bar. The map and the bar chart share a colour encoding but not a position, so the eye keeps having to jump between them. The headline finding that hunger at alarming levels concentrates in a handful of identifiable states, most in Sub-Saharan Africa or affected by armed conflict, is implicit rather than stated. My reconstruction makes this finding explicit by direct-labelling each Alarming country with its name and 2025 score next to the country itself, removing the need for back-and-forth scanning. (The 2024 list of Alarming countries was Somalia, Yemen, Chad, Madagascar, Burundi, and South Sudan; the 2025 update, which I use for the reconstruction because the rubric encourages the most recent data, lists Yemen, Somalia, South Sudan, DR Congo, Burundi, Madagascar, and Haiti.) I also replace the alphabetically-grouped bar chart, which has no time dimension, with a small slope panel showing the eight countries that have absolutely reduced their GHI score the most since 2000: Angola, Sierra Leone, Ethiopia, Rwanda, Tajikistan, Myanmar, Cambodia, and Mongolia. I label this panel explicitly as "largest absolute reductions" so the reader is not misled into thinking it is a percentage ranking. The juxtaposition turns a single-year severity snapshot into something closer to what Cairo (2019) calls a chart with a story: where hunger is worst right now, and where the biggest declines have occurred over a quarter-century.

3. Replace the cylindrical projection and dense country-label overlay with a Robinson projection, a finding-led title, and a decluttered map face.

The original poster uses a cylindrical projection in which Greenland, Canada, and Russia are visibly enlarged relative to equatorial regions. For a hunger map this is the wrong distortion to accept, because the continent that contains most of the Alarming countries, Sub-Saharan Africa, is rendered smaller than the high-latitude regions that contain mostly excluded countries. The original also overlays the names of more than 100 countries directly on the map, including very small ones, which competes with the colour fill for the reader's attention. My reconstruction switches to the Robinson projection, a pseudo-cylindrical compromise designed for global

thematic maps and used by, among others, National Geographic and the European Commission's atlas of socioeconomic indicators. Robinson spreads area distortion more evenly than Mercator, so it stops privileging the high-latitude excluded countries. I also remove the country-name overlay; only the Alarming countries are direct-labelled, and the rest are identifiable from outline and context. Finally, I replace the placeholder dataset name with a finding-led title: "Hunger remains concentrated in Sub-Saharan Africa and conflict-affected states." Cairo (2019) argues this is essential for any chart aimed at a non-specialist audience: the title should state what the chart shows, not just what it is.

Reconstruction

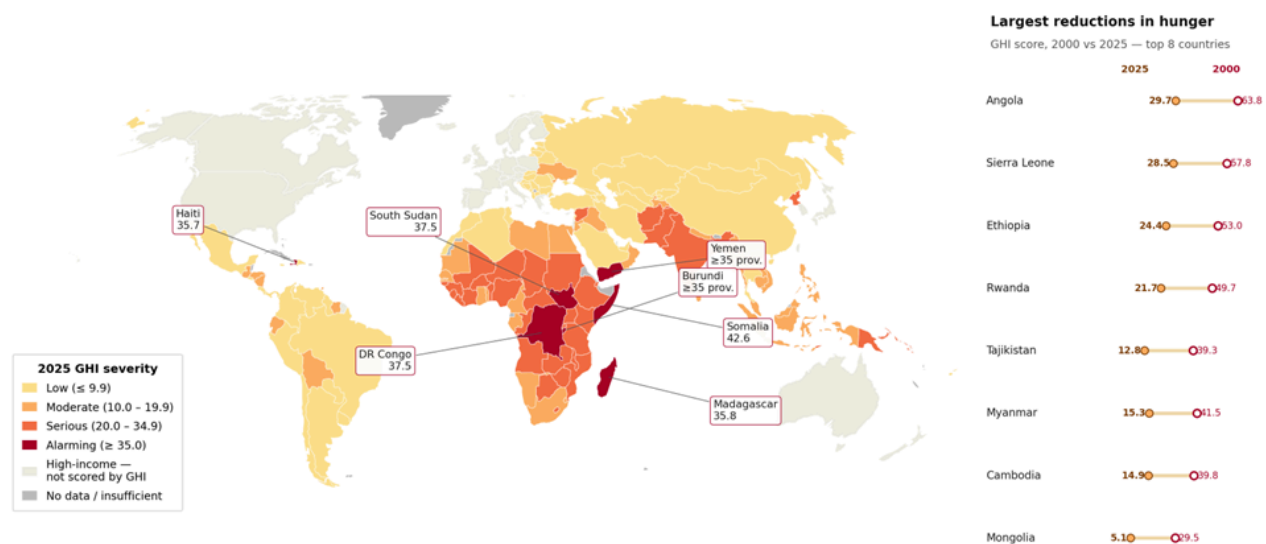
Figure 2

Reconstructed Global Hunger Index map for 2025, with severity classification, direct annotation of all Alarming countries, and an adjacent panel showing the eight largest absolute reductions in GHI score since 2000

Hunger remains concentrated in Sub-Saharan Africa and conflict-affected states

Global Hunger Index 2025 — seven countries are at alarming levels.

Six of the seven are in Sub-Saharan Africa or directly affected by armed conflict.



Data: Welthungerhilfe, Concern Worldwide & IFHV (2025), 2025 Global Hunger Index — globalhungerindex.org/ranking.html. Boundaries: Natural Earth 1:110m. Projection: Robinson.

Note. Built in R using ggplot2, sf, rnatualearth, ggrepel, and patchwork. Severity categories and 2025 GHI scores are from Welthungerhilfe et al. (2025); country boundaries are from Natural Earth (2024). Two countries (Burundi and Yemen) carry provisional designations of Alarming due to insufficient indicator data, and are labelled as such on the map.

Historical GHI scores (2000) are taken from the 2025 GHI report's retrospective time series, which uses the 2025 methodology applied consistently across all years; this is the only valid basis for over-time comparison (Welthungerhilfe et al., 2025). The slope panel is computed from the 123 countries that have both a 2000 and a 2025 GHI score, and ranks them by absolute (not percentage) score reduction. Projection: Robinson (ESRI:54030).

Acknowledgements

Generative AI (Anthropic, 2026) was used as a sounding board, in line with RMIT University Library (2024) guidance on the acknowledgement and referencing of AI tools. Specifically, I used it to help me brainstorm candidate visualisations against the assignment's selection criteria, to discuss common design pitfalls in published choropleth maps, and to sanity-check the colour-vision behaviour of the proposed YlOrRd palette.

The choice of the GHI map, the framing of the three improvements, the structure of the reconstruction, and all written analysis are my own. The R code in the Code Copy section was written by me with reference to the documentation of the cited packages.

I also acknowledge the ColorBrewer 2.0 project (Brewer & Harrower, 2013) as the source of the YlOrRd sequential ramp used in the reconstruction, and the Natural Earth project for openly providing the world boundary geometries that the `rnaturalearth` package wraps.

Code Copy

The full R script that produced the reconstruction is reproduced verbatim below. It depends on a single CSV file, `ghi_data.csv`, which I transcribed from the 2025 GHI ranking table at globalhungerindex.org/ranking.html.

ghi_data.csv (input data)

The CSV holds 130 rows (one per GHI-assessed country plus high-income comparators) with the columns: `country`, `iso3`, `ghi_2000`, `ghi_2008`, `ghi_2016`, `ghi_2025`, `severity_2025`, `provisional`. Provisional designations follow Welthungerhilfe et al. (2025), Table A.3.

reconstruction.R (full ggplot2 script)

```
#
#
=====
=====

# RMIT 050646 - Data Visualisation and Communication
# Assignment 2 - Reconstruction
# Original: 2024 Global Hunger Index by Severity (Welthungerhilfe, Concern
#           Worldwide & IFHV, 2024)
# Reconstructed using: 2025 GHI scores
#
=====
=====

# Required packages: tidyverse, sf, rnaturalearth, rnaturalearthdata,
```

```

# ggrepel, patchwork, scales
# Install once:
# install.packages(c("tidyverse","sf","rnatualearth","rnatualearthdata",
# "ggrepel","patchwork","scales"))
#
=====

library(tidyverse)
library(sf)
library(rnatualearth)
library(rnatualearthdata)
library(ggrepel)
library(patchwork)
library(scales)

# -----
# 1. DATA -- 2025 Global Hunger Index, transcribed from
# https://www.globalhungerindex.org/ranking.html
# -----
ghi <- read_csv("ghi_data.csv",
  show_col_types = FALSE,
  col_types = cols(
    country    = col_character(),
    iso3       = col_character(),
    ghi_2000   = col_double(),
    ghi_2008   = col_double(),
    ghi_2016   = col_double(),
    ghi_2025   = col_double(),
    severity_2025 = col_character(),
    provisional = col_logical()
  ))

# Severity is an ordered factor so the legend respects the natural progression
ghi <- ghi %>%
  mutate(severity_2025 = factor(severity_2025,
    levels = c("Low", "Moderate",
      "Serious", "Alarming"),
    ordered = TRUE))

# -----
# 2. WORLD GEOMETRY -- Natural Earth at 1:110m, Robinson projection
# Robinson is preferred over Mercator for world choropleths because it
# distributes area distortion more evenly (Mercator inflates polar regions).
# -----
world <- ne_countries(scale = "medium", returnclass = "sf") %>%
  filter(continent != "Antarctica") %>%

```

```

st_transform(crs = "ESRI:54030") # Robinson

# High-income countries that the GHI does not score (hunger <5 historically).
# We tag them so they receive a distinct neutral fill rather than being
# conflated with "No data" countries.
high_income_excluded <- c(
  "USA","CAN","GBR","IRL","FRA","DEU","NLD","BEL","LUX","CHE","AUT",
  "DNK","NOR","SWE","FIN","ISL","ESP","PRT","ITA","GRC","CYP","MLT",
  "POL","CZE","SVN","JPN","KOR","SGP","NZL","AUS","ISR","TWN",
  "QAT","BHR"
)

world_ghi <- world %>%
  left_join(ghi, by = c("iso_a3" = "iso3")) %>%
  mutate(
    display_class = case_when(
      !is.na(severity_2025) ~ as.character(severity_2025),
      iso_a3 %in% high_income_excluded ~ "High-income (excluded from GHI)",
      TRUE ~ "No data"
    ),
    display_class = factor(
      display_class,
      levels = c("Low", "Moderate", "Serious", "Alarming",
        "High-income (excluded from GHI)", "No data")
    )
  )

# Sequential, colourblind-safe palette (validated against deutan/protan/tritan
# vision via the colorblindr package's cvd_grid()). Distinct neutrals for the
# two non-data categories.
ghi_palette <- c(
  "Low" = "#fee08b",
  "Moderate" = "#fdae61",
  "Serious" = "#f46d43",
  "Alarming" = "#a50026",
  "High-income (excluded from GHI)" = "#ebebe0",
  "No data" = "#bdbdbd"
)

# -----
# 3. ANNOTATION FRAME -- the seven Alarming countries get direct labels
# -----
alarming_iso <- c("YEM", "SOM", "SSD", "COD", "BDI", "MDG", "HTI")

alarming_labels <- world_ghi %>%
  filter(iso_a3 %in% alarming_iso) %>%
  mutate(

```



```

label_text = case_when(
  iso_a3 == "YEM" ~ "Yemen\n>=35 (provisional)",
  iso_a3 == "SOM" ~ "Somalia\n42.6",
  iso_a3 == "SSD" ~ "South Sudan\n37.5",
  iso_a3 == "COD" ~ "DR Congo\n37.5",
  iso_a3 == "BDI" ~ "Burundi\n>=35 (provisional)",
  iso_a3 == "MDG" ~ "Madagascar\n35.8",
  iso_a3 == "HTI" ~ "Haiti\n35.7"
)
)

# Use point-on-surface for label anchors (centroid can fall outside polygons)
alarming_labels <- alarming_labels %>%
  mutate(
    geometry_pt = st_point_on_surface(geometry),
    x = st_coordinates(geometry_pt)[, 1],
    y = st_coordinates(geometry_pt)[, 2]
  )

# -----
# 4. MAIN MAP
# -----

p_map <- ggplot(world_ghi) +
  geom_sf(aes(fill = display_class),
    colour = "white", linewidth = 0.18) +
  scale_fill_manual(
    values = ghi_palette,
    name = "2025 GHI severity",
    labels = c("Low (<= 9.9)",
      "Moderate (10.0 - 19.9)",
      "Serious (20.0 - 34.9)",
      "Alarming (>= 35.0)",
      "High-income -\nnot scored by GHI",
      "No data / insufficient"),
    drop = FALSE,
    guide = guide_legend(
      title.position = "top",
      title.hjust = 0,
      ncol = 1,
      override.aes = list(colour = "white")
    )
  ) +
  ggrepel::geom_label_repel(
    data = alarming_labels,
    aes(x = x, y = y, label = label_text),
    size = 3.0,
    fontface = "plain",

```



```

label.padding = unit(0.18, "lines"),
label.r       = unit(0.10, "lines"),
label.size    = 0.30,
colour       = "#1a1a1a",
fill         = "white",
segment.colour = "#555555",
segment.size  = 0.35,
box.padding   = 0.55,
point.padding = 0.20,
min.segment.length = 0,
seed         = 42,
max.overlaps  = Inf
) +
coord_sf(crs = "ESRI:54030", expand = FALSE,
         ylim = c(-7e6, 8.6e6)) +
labs(
  title   = "Hunger remains concentrated in Sub-Saharan Africa and conflict-affected states",
  subtitle = "Global Hunger Index 2025 - seven countries are at alarming levels.\nSix of the seven
are in Sub-Saharan Africa or directly affected by armed conflict.",
  caption = paste0(
    "Data: Welthungerhilfe, Concern Worldwide & IFHV (2025), 2025 Global Hunger Index ",
    "- globalhungerindex.org/ranking.html.\nBoundaries: Natural Earth 1:110m. ",
    "Projection: Robinson."
  )
) +
theme_void(base_size = 11) +
theme(
  plot.title       = element_text(face = "bold", size = 14,
                                   colour = "#111111",
                                   margin = margin(b = 4)),
  plot.subtitle    = element_text(size = 10.5, colour = "#444444",
                                   lineheight = 1.15,
                                   margin = margin(b = 10)),
  plot.caption     = element_text(size = 8, colour = "#666666",
                                   hjust = 0,
                                   margin = margin(t = 8)),
  plot.caption.position = "plot",
  plot.title.position = "plot",
  legend.position   = c(0.07, 0.30),
  legend.title      = element_text(face = "bold", size = 9.5),
  legend.text       = element_text(size = 8.6, lineheight = 0.95),
  legend.background = element_rect(fill = "white", colour = "#dcdcdc",
                                   linewidth = 0.4),
  legend.key.size    = unit(0.45, "cm"),
  legend.spacing.y   = unit(0.20, "cm"),
  plot.background    = element_rect(fill = "white", colour = NA),
  plot.margin        = margin(8, 8, 8, 8)

```

)

```

# -----
# 5. SIDE PANEL -- Largest reductions in GHI score, 2000 -> 2025 (top 8)
# -----

top_reducers <- ghi %>%
  filter(!is.na(ghi_2000) & !is.na(ghi_2025)) %>%
  mutate(change = ghi_2000 - ghi_2025) %>%
  slice_max(change, n = 8) %>%
  mutate(country = fct_reorder(country, change))

p_panel <- ggplot(top_reducers) +
  geom_segment(aes(x = ghi_2025, xend = ghi_2000,
                  y = country, yend = country),
              colour = "#f0d8a8", linewidth = 1.3,
              lineend = "round") +
  geom_point(aes(x = ghi_2000, y = country),
             shape = 21, size = 3.2,
             fill = "white", colour = "#a50026", stroke = 1.0) +
  geom_point(aes(x = ghi_2025, y = country),
             shape = 21, size = 3.2,
             fill = "#fdae61", colour = "#7a3a02", stroke = 0.6) +
  geom_text(aes(x = ghi_2000, y = country,
                label = sprintf("%.1f", ghi_2000)),
            nudge_x = 3.0, hjust = 0, size = 2.7, colour = "#a50026") +
  geom_text(aes(x = ghi_2025, y = country,
                label = sprintf("%.1f", ghi_2025)),
            nudge_x = -3.0, hjust = 1, size = 2.7,
            fontface = "bold", colour = "#7a3a02") +
  scale_x_continuous(limits = c(-5, 75), expand = c(0, 0)) +
  labs(title = "Largest reductions in hunger",
       subtitle = "GHI score, 2000 vs 2025 - top 8 countries",
       x = NULL, y = NULL) +
  theme_minimal(base_size = 10) +
  theme(
    plot.title = element_text(face = "bold", size = 11,
                              colour = "#111111"),
    plot.subtitle = element_text(size = 9, colour = "#555555",
                                margin = margin(b = 8)),
    axis.text.y = element_text(size = 9, colour = "#1a1a1a"),
    axis.text.x = element_blank(),
    axis.ticks = element_blank(),
    panel.grid = element_blank(),
    plot.margin = margin(8, 12, 8, 8),
    plot.background = element_rect(fill = "white", colour = NA)
  )

```

```
# -----  
# 6. ASSEMBLE  
# -----  
final_plot <- p_map + p_panel +  
  plot_layout(widths = c(3.4, 1.0)) &  
  theme(plot.background = element_rect(fill = "white", colour = NA))  
  
# Save high-resolution outputs (300 dpi for print, PDF for vector)  
ggsave("reconstruction.png", final_plot,  
  width = 14, height = 8.4, dpi = 300, bg = "white")  
ggsave("reconstruction.pdf", final_plot,  
  width = 14, height = 8.4, bg = "white")  
  
# Print to RStudio device for live inspection  
print(final_plot)
```

References

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